

The acute effect of green tea consumption on endothelial function in healthy individuals.

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BACKGROUND: Tea consumption is associated with decreased cardiovascular risk. Flow-mediated dilatation (FMD) of the brachial artery is related to coronary endothelial function and it is an independent predictor of cardiovascular risk. Black tea has a beneficial effect on endothelial function; the effect, however, of green tea on brachial artery reactivity has not been defined yet. **DESIGN AND METHODS:** We studied 14 healthy individuals (age 30 $\pm$ 3 years) with no cardiovascular risk factors except from smoking (50%) on three separate occasions on which they took: (a) 6 g of green tea, (b) 125 mg of caffeine (the amount contained in 6 g of tea), or (c) hot water. FMD of the brachial artery was measured before each intervention and 30, 90, and 120 min afterward. High-sensitivity C-reactive protein, interleukins 6 (IL-6) and 1b (IL-1b), total plasma antioxidative capacity, and total plasma oxidative status/stress were measured at baseline and at 120 min after each intervention. **RESULTS:** Resting and hyperemic brachial artery diameter did not change either with tea or with caffeine. FMD increased significantly with tea (by 3.69%, peak at 30 min, $P<0.02$), whereas it did not change significantly with caffeine (increase by 1.72%, peak at 30 min, $P=NS$). Neither tea nor caffeine had any effect on high-sensitivity C-reactive protein, IL-6, IL-1b, total plasma antioxidative capacity, or total plasma oxidative status/stress. **CONCLUSION:** Green tea consumption has an acute beneficial effect on endothelial function, assessed with FMD of the brachial artery, in healthy individuals. This may be involved in the beneficial effect of tea on cardiovascular risk.